Gruithuisen Data

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0.1 Gruithuisen Data

- Historical manuscripts by Franz von Paula Gruithuisen (Munich Observatory, early 19th century) containing daily notes on weather, sky conditions, and sunspot observations, written in Kurrent (old German script).
- The notebooks combine textual remarks ("Flecken", "Öffnungen") with drawings — both small in-text sketches and large full-disk solar diagrams.
- Working data product: daily entries with date, time, presence/absence of spots, NG/NS counts (if visible), drawing type/quality flag, weather/seeing notes, and page references.

0.1.1 Importance (for FARSuN & SN V3)

- Provides dense coverage during the late Dalton Minimum (~1817 – mid-1840s), filling a critical gap in early-19th-century solar records.
- Includes both spotless and active days — essential for long-term continuity and low-activity baselines in the Sunspot Number (SN V3).

- Weather and seeing information enrich the FARSuN condition table, supporting observer-bias and visibility studies.
- Enables cross-calibration with contemporary observers (Adams, Stark, Pastorff) to refine k-factors and reliability estimates.

0.1.2 Time Span (working range)

- Observations extend roughly 1817 – 1849 (core dense years 1824–1826 pending verification).
- Task under way to finalize start/end years and compile a full volume + page inventory.

0.1.3 What is being extracted

- Core daily parameters: date, time, spotless/active flag, NG, NS, comments, and provenance (text vs drawing).
- Drawing flags: schema / full-disk (indicator of counting fidelity).
- Contextual data: sky condition, barometer, hygrometer, temperature, method remarks.
- Metadata links: book / volume / page numbers, normalized image IDs, checksums.

0.2 Current Status (Oct 2025)

- Data retrieval: All image scans and text snippets collected; preliminary CSV/XLSX files prepared.
- Digitization: Page crops and snippet images completed; Transkribus/Cloude App workflow/pipeline is prepared for Kurrent script.
- Counting: Double-counting verification procedure definition in progress.
- Quality control: Two-pass QC (survey → counting) in progress;
- Metadata: Observer/site/instrument registry being merged with database; VO/EPN-TAP export planned after QC completion.

0.2.1 Quality Control & Normalization

- Enforce standardized filenames, date/time formats, and provenance flags.
- Maintain audit trail linking every image to its extracted record.
- Ambiguous or illegible Kurrent passages flagged for expert review; define acceptance thresholds for HTR accuracy.

0.2.2 Open Items

• Formalize counting policy for small in-text sketches (presence-only vs countable). • Benchmark Transkribus/App HTR results; set minimum human-review rate. • Identify overlaps with coeval observers for calibration validation.

0.3 Next Steps (toward SN V3 integration)

- Complete survey pass → definitive daily presence/absence + drawing availability/type.
- Generate provisional daily NG/NS with uncertainty and provenance flags.
- Apply QC (dual counts, normalization, conflict resolution, metadata registry).
- Perform inter-observer calibration and bias checks.
- Publish FAIR, VO-ready tables (daily counts + condition data + image catalog) via ROB/SILSO.

Gruithuisen's manuscripts offer a long, context-rich, early-19th-century record of solar and meteorological activity. With extraction, QC nearly complete, this dataset is poised to become a cornerstone for SN V3 calibration and uncertainty modeling within the FARSuN project.